

Matrix Multiplication: The Core Operation

In linear algebra, **matrix multiplication** is a binary operation that produces a matrix from two matrices. For matrix multiplication, the number of **columns** in the first matrix must be equal to the number of **rows** in the second matrix.

The resulting matrix, known as the **matrix product**, has the number of rows of the first and the number of columns of the second matrix.

The product of matrices **A** and **B** is denoted as **AB**.

This is how matrix multiplication works:

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1x_1 + 2x_2 \\ 3x_1 + 4x_2 \end{bmatrix}$$

Matrix : An Example

Example:

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}$$

The output size comes from the **outer dimensions**:

$$2 \times 2 \rightarrow 2 \times 2$$

Every output value is a dot product.

To calculate C_{ij} :

Take row i from A $\rightarrow$ take column j from B $\rightarrow$ multiply corresponding values $\rightarrow$ add them.

For example:

$$C_{11} = (1 \times 5) + (2 \times 7) = 19$$

$$C_{12} = (1 \times 6) + (2 \times 8) = 22$$

Row $\times$ Column $\rightarrow$ One Number

Repeat this for every position in the output matrix.

Why AI Needs Matrix Multiplication?

A neural network represents:

Input data → Matrix

Weights → Matrix

Bias → Vector

Output → Matrix/Vector

A basic neural-network layer performs:

$$Y = XW + b$$

Example:

Input: 5,000 customer records × 20 features

Weights: 20 × 64

Output:

$$5000 \times 20 \times 20 \times 64$$

→ **5,000 × 64**

That single matrix multiplication transforms the original features into **64 learned representations**.

The same fundamental operation appears throughout **neural networks, CNNs, transformers and embeddings**.

Resources To Learn Further

3Blue1Brown — Essence of Linear Algebra

Best for visualized working of matrices.

MIT OpenCourseWare — Linear Algebra

More rigorous mathematical foundation with complex examples.

Khan Academy — Matrix Multiplication

Beginner-friendly practice and detailed explanations.

DeepLearning.AI — Mathematics for Machine Learning

Connects linear algebra directly to machine learning.